International Institute Of Information Technology Bangalore

AIM-102

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Statistical Machine Learning
Course Project

1 Introduction

Natural Language Processing (NLP) is widely used for analyzing textual data. In this project, we compare classical machine learning models using two different feature representations: TF-IDF (sparse features) and contextual embeddings generated using DistilBERT.

The objective is to analyze how feature representation affects classification performance in sentiment analysis.

2 Dataset Description and EDA

We use the IMDb dataset from Hugging Face, which is a binary classification dataset for sentiment analysis.

- Number of samples used: 2000 (subset)
- Classes: Positive and Negative

Basic exploratory data analysis (EDA) was performed, including class distribution and text length analysis.

3 Methodology

3.1 TF-IDF Features

Text preprocessing included cleaning and tokenization. TF-IDF vectorization was used to convert text into numerical feature vectors.

3.2 Embedding Features

We used DistilBERT as a pretrained encoder model to extract embeddings. The model was used strictly as a feature extractor without fine-tuning.

Mean pooling was applied to obtain sentence-level embeddings.

3.3 Models

The following classical models were used:

- Logistic Regression
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- KMeans Clustering

4 Experiments

Experiments were conducted on both TF-IDF and embedding-based features. PCA was applied for dimensionality reduction, and t-SNE was used for visualization.

5 Results

5.1 Performance Comparison

Feature	Model	Accuracy	F1
TF-IDF	Logistic	0.8756	0.8766
TF-IDF	SVM	0.8618	0.8617
TF-IDF	KNN	0.6775	0.6800
TF-IDF	KMeans	0.5762	-
Embeddings	Logistic	0.8105	0.8125
Embeddings	SVM	0.7895	0.7927
Embeddings	KNN	0.7250	0.6998
Embeddings	KMeans	0.5410	-

5.2 PCA Visualization

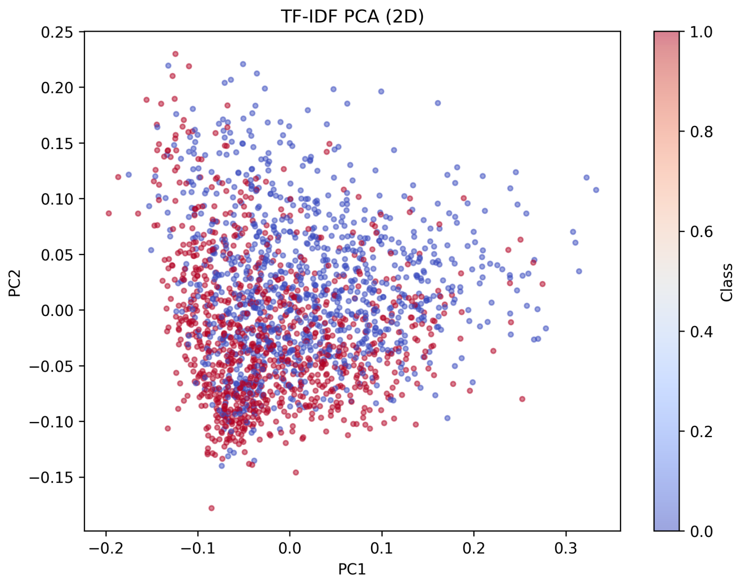

Figure 1: PCA Visualization for TF-IDF

5.3 t-SNE Visualization

6 Discussion

From the results, TF-IDF features outperform embedding-based features across most models. Logistic Regression achieves the highest accuracy of 0.8756 with TF-IDF.

This is likely because TF-IDF is highly effective for sentiment analysis tasks, where word presence and frequency are strong indicators of class labels. In contrast, embeddings capture semantic relationships but may not emphasize key sentiment-bearing words as effectively in a frozen setting.

KNN performs poorly on TF-IDF due to high dimensionality, while it performs slightly better on embeddings due to lower dimensional representation.

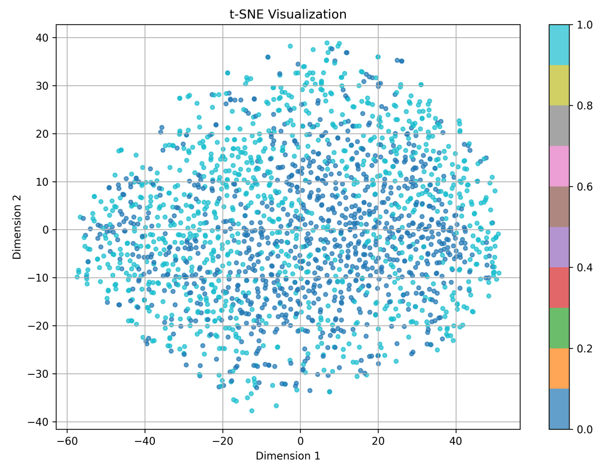

Figure 2: t-SNE Visualization for Embeddings

KMeans shows the lowest performance, as it is an unsupervised algorithm and does not directly optimize classification objectives. Cluster-to-label mapping introduces additional uncertainty, and hence F1-score is not reported.

PCA helps reduce dimensionality and noise, but excessive reduction can lead to loss of important information. t-SNE visualizations indicate that embedding features form smoother clusters, although TF-IDF provides better classification performance.

7 Conclusion

In this project, we compared classical machine learning models using TF-IDF and contextual embeddings. TF-IDF features achieved higher accuracy for this task, demonstrating their effectiveness in sentiment classification.

Embedding-based approaches provide richer semantic representations but may require fine-tuning to outperform traditional methods.

8 Learnings

- Importance of feature representation in NLP tasks
- Trade-offs between sparse and dense representations
- Benefits of batching and GPU acceleration
- Insights from dimensionality reduction techniques